EC3005 — Banking and Finance

Winter 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC3005 — Banking and Finance
Lecturer	Dr. Fergal O'Connor
Institution	University College Cork
Session	Winter 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions. Attempt **2 of 3**. Each question weighted equally (100 marks each, 100% of the paper).
Materials	Non-programmable calculator permitted

Coverage note. This scheme works through **all three** Section A questions (A1, A2, A3) in full, since you won't know in advance which two the exam will ask you to answer under exam conditions — study all three and pick your strongest pair on the day.

Own-figure rule. Where a question part carries forward from an earlier numerical or conceptual step, an answer that applies the correct method consistently from an earlier error still earns the marks available for the later steps. Markers (and you, self-marking) should apply this throughout the numerical parts of A3.

Source basis. Content and worked methods follow Dr. O'Connor's own EC3005 course pack and lecture notes (financial intermediation, financial structure, regulation, bonds, and the module's own bond-price / stock-price / future-value formula sheet reproduced with this paper), cross-checked against the module's core texts — Mishkin, *The Economics of Money, Banking and Financial Markets*, and Bodie, Kane & Marcus, *Investments* — for the standard treatment of any concept not spelled out in the lecture notes. Where lecturer-specific terminology could differ from the standard textbook framing (e.g. exactly which four tools are presented for the moral hazard problem), that is flagged at the relevant question.

Section A · Question 1

Financial markets, fractional reserve banking, financial intermediation — 100 marks

A1(a) [30 marks]

Financial markets can be classified as primary and secondary markets. Outline the difference between primary and secondary markets.

A1(b) [30 marks]

Explain the idea that Fractional Reserve Banking increases the money supply.

A1(c) [40 marks]

Financial intermediaries and markets can alleviate market frictions and contribute to economic growth by: (i) producing information and allocating capital; (ii) monitoring firms and exerting corporate governance; (iii) mobilising/pooling savings; (iv) easing the exchange of goods and services. Discuss **any two** of the above functions and comment on how each function may contribute to economic growth.

FULL MODEL ANSWER

Primary market: the market in which a security is sold **for the first time** — the issuer (a company issuing new shares, or a government issuing new bonds) receives the proceeds directly. An IPO and a government bond auction are both primary-market transactions.

Secondary market: the market in which securities that have **already been issued** are bought and sold between investors. The issuer is not a party to the trade and receives no funds from it — the New York Stock Exchange and the Irish Stock Exchange (Euronext Dublin) are secondary markets.

Why this matters: the two are functionally linked, not just definitionally different. A deep, liquid secondary market is what makes investors willing to buy in the primary market in the first place — an investor who could not resell a bond or share easily would demand a much higher return to compensate for that illiquidity, raising the cost of raising finance for issuers. Secondary-market liquidity is therefore a precondition for a well-functioning primary market, not a separate phenomenon.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	Correct distinction stated explicitly in terms of **who receives the funds** (issuer vs. no one — trade is between investors) **AND** at least one concrete example of each market **AND** the liquidity link between the two explained.
15–22	**15–22**	Correct distinction with at least one example, but the liquidity/pricing link between primary and secondary markets is not explained, or is asserted without reasoning.
8–14	**8–14**	Only one of the two markets is correctly defined, or the distinction is stated vaguely (e.g. "new vs old shares") without reference to who receives the funds.
0–7	**0–7**	No meaningful distinction made, or primary/secondary is confused with an unrelated classification (e.g. money market vs. capital market).

> **Mechanism:** when a bank receives a deposit, it is required to hold only a fraction of it as reserves (the **reserve ratio**) and is free to lend out the rest. The borrower spends that loan, and whoever receives the payment deposits it at a bank (possibly the same one, possibly another) — which again holds back only the reserve ratio and lends out the remainder. Each round of lending is smaller than the last, shrinking geometrically by a factor of **(1 − reserve ratio)**, but the process does not stop after one round: it repeats indefinitely, and it is the sum of every round that determines how much the money supply actually expands.

Deposit Expansion Under Fractional Reserve Banking

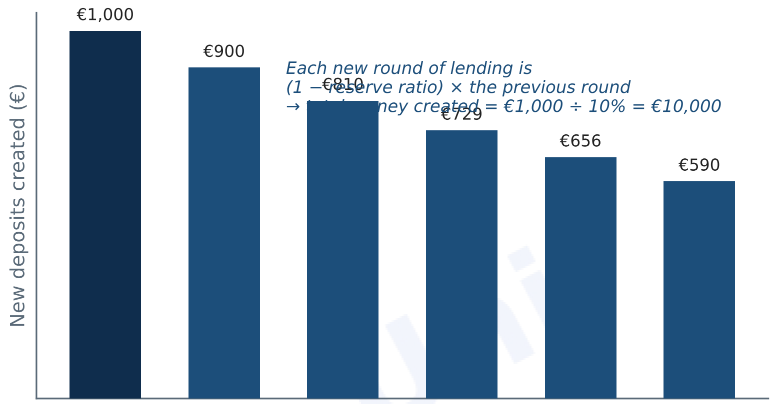

Fig. 1 — Successive rounds of deposit expansion under fractional reserve banking (illustrative: €1,000 initial deposit, 10% reserve ratio), shrinking geometrically and summing to the total money created.

Round	New deposit created (€)	Held as reserves (€)	Re-lent (€)
1	1,000.00	100.00	900.00
2	900.00	90.00	810.00
3	810.00	81.00	729.00
4	729.00	72.90	656.10
…	**(geometric series continues)**		
Total money created	**€10,000.00**		**= €1,000 ÷ 10%**

> **Result:** summing the full geometric series of new deposits gives total money

created = initial deposit ÷ reserve ratio. With a 10% reserve ratio, the **money multiplier** is 1/0.10 = **10**, so a €1,000 initial deposit ultimately supports €10,000 of money in the system — nine times the original deposit was *created* through repeated lending, none of it printed by a central bank.

EXAM TECHNIQUE

The reserve ratio used above (10%) is illustrative — the exact multiplier depends on the actual required reserve ratio in force, and in practice the theoretical multiplier is never fully reached because some cash leaks out of the banking system into public holdings and banks may choose to hold excess reserves beyond the legal minimum. State the formula (1/reserve ratio) and show the geometric reasoning — a numeric example with any sensible reserve ratio is sufficient to earn full marks, the specific ratio chosen is not being tested.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	Correctly explains that banks lend out deposits net of required reserves **AND** shows that this repeats across successive rounds of re-depositing/re-lending **AND** gives the money multiplier (1/reserve ratio) with a correct worked numeric example.
15–22	**15–22**	Correctly explains one round of lending increasing the money supply, but does not extend the reasoning to successive rounds, or states the multiplier formula without a worked example.
8–14	**8–14**	Vague reference to "banks lending out deposits" without reference to reserve requirements or any multiplier effect.
0–7	**0–7**	No credible mechanism given, or confuses fractional reserve banking with an unrelated concept (e.g. quantitative easing).

EXAM TECHNIQUE

The question asks for **any two** of the four functions listed — answering all four does not earn extra marks and risks running out of time relative to the depth expected per function. This answer covers (i) producing information and allocating capital and (ii) monitoring firms and exerting corporate governance; either of the other two pairings is equally valid if argued with the same depth.

Producing information and allocating capital: individual savers face high costs in researching which borrowers are creditworthy — this is exactly the kind of task with large fixed costs and economies of scale, which is why it is delegated to specialist intermediaries rather than done by each saver separately. Banks and other intermediaries develop the expertise to screen borrowers before lending (**reducing adverse selection**) and monitor them afterwards. **Growth link:** by channelling savings toward the projects with the highest genuine returns rather than the projects that merely look best on paper, intermediaries raise the average productivity of the economy's capital stock — capital is allocated to its most productive use rather than spread inefficiently across borrowers who cannot be properly screened by individual savers.

Monitoring firms and exerting corporate governance: once a loan or investment is made, the intermediary has an ongoing incentive to monitor the firm's use of funds (**reducing moral hazard**) — a bank can attach loan covenants, and equity intermediaries such as venture capital and private equity funds routinely take board seats. **Growth link:** this disciplines management toward value-maximising decisions and away from empire-building or shirking, which — aggregated across the whole economy — raises the return firms generate on invested capital and therefore the rate at which the capital stock itself grows. (See also Q. A2(c), where equity monitoring is examined specifically as a tool against moral hazard.)

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31–40	31–40	Both chosen functions correctly defined **AND** a specific, non-generic growth mechanism given for each (not just "this helps the economy") **AND** the underlying market friction (adverse selection/moral hazard/coordination cost) each function addresses is named.
21–30	21–30	Both functions correctly defined with a growth link stated, but the growth link is generic/asserted rather than mechanistically explained, or the underlying friction is not identified.
11–20	11–20	Only one function is developed to the standard above; the second is defined but not linked to growth, or is materially incomplete.
0–10	0–10	Neither function is developed beyond a one-line definition, or fewer than two functions are meaningfully addressed.

Question A1 Total: 100 Marks

Section A · Question 2

Shadow banking, bank regulation, moral hazard in equity contracts — 100 marks

A2(a) [30 marks]

Explain how financial innovation led to the growth of the shadow banking system.

A2(b) [30 marks]

Outline the economic rationale for bank regulation. Discuss the difference between macro-prudential and micro-prudential bank regulation.

A2(c) [40 marks]

Discuss four tools that help solve the moral hazard problem in equity contracts.

FULL MODEL ANSWER

Mechanism: financial innovation — chiefly **securitisation** — allowed banks to pool loans (mortgages, auto loans, credit-card receivables) into a special-purpose vehicle and sell claims on the pool's cash flows to investors as assetbacked securities, rather than holding the loans on their own balance sheet until maturity. This is the **"originate-to-distribute"** model: the originating bank earns a fee for making and packaging the loan but no longer bears its credit risk, which is passed on to the investors who buy the securitised claim.

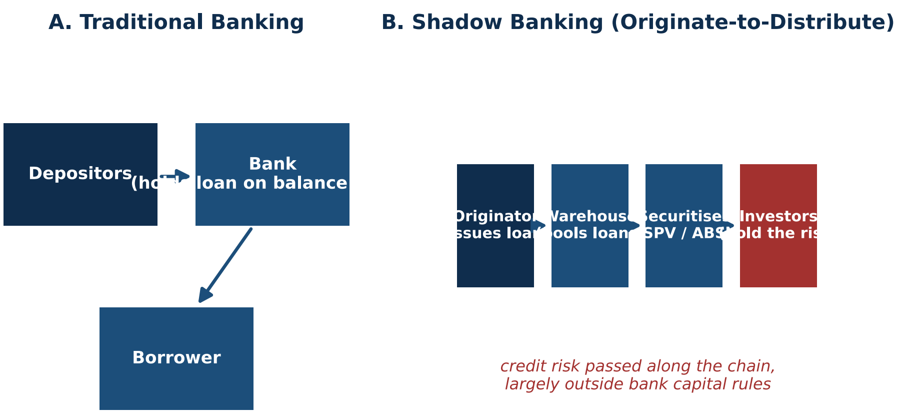

Fig. 2 — Traditional bank lending, where the originating bank holds the loan and its credit risk, compared with the shadow-banking originate-to-distribute chain, where the risk is passed along to investors outside the regulated banking system.

Growth of shadow banking: because the entities performing each stage of this chain — money-market funds, investment banks, structured investment vehicles — undertake credit intermediation (borrowing short, lending/investing long) **without** taking insured deposits, they fell largely outside traditional bank capital and liquidity regulation. This let credit intermediation expand well beyond what banks' own balance sheets and capital requirements would otherwise have permitted, since risk could be moved off-balance-sheet as fast as new loans were originated.

EXAM TECHNIQUE

A common error is describing securitisation mechanically without stating *why* it produced growth in shadow banking specifically — the answer must connect the innovation to the fact that shadow banking entities sit outside bank regulation. The regulatory gap this exposed is exactly what macro-prudential regulation (see Q. A2(b)) was extended after 2008 to try to close.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	Explains securitisation/originate-to-distribute correctly **AND** explains that shadow banking entities sit outside traditional bank capital/liquidity regulation **AND** connects this explicitly to why credit intermediation could expand faster than under the traditional model.
15–22	**15–22**	Correctly describes securitisation but does not explain why this let shadow banking grow specifically (the regulatory-gap point is missing or only implied).
8–14	**8–14**	Vague reference to "financial innovation" or "complex products" without naming securitisation or the originate-to-distribute mechanism.
0–7	**0–7**	No credible mechanism given, or shadow banking is conflated with ordinary bank lending.

Economic rationale for regulation: left alone, banking has three features that justify intervention beyond what applies to an ordinary firm: (i) **information asymmetry** between depositors and banks about asset quality, which can trigger self-fulfilling bank runs even at solvent institutions; (ii) **negative externalities** from failure — one bank's collapse can spread through interbank exposures and confidence effects to the rest of the financial system (**systemic risk**), a cost the failing bank does not fully internalise; and (iii) **moral hazard from deposit insurance** itself — insured depositors have little incentive to monitor bank risk-taking, which can encourage banks to take on more risk than they otherwise would.

	Micro-prudential regulation	Macro-prudential regulation
Focus	The safety and soundness of an **individual** institution	The stability of the **financial system as a whole**
Objective	Protect depositors/investors at that firm; limit its idiosyncratic risk	Limit systemic risk and the build-up of system-wide vulnerabilities
Typical tools	Minimum capital ratios, firm-level supervision, deposit insurance	Countercyclical capital buffers, systemic risk buffers, stress tests of the system
Treats risk as...	Exogenous to the system (independent of other firms' behaviour)	Endogenous — driven partly by the collective behaviour of all institutions together

Why the distinction matters: a bank can be individually well-capitalised by micro-prudential standards and still contribute to system-wide fragility if many banks are simultaneously exposed to the same risk (e.g. a common real-estate downturn) — this is precisely the gap macro-prudential regulation was developed to close after the 2007–08 crisis exposed it.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	At least two distinct rationales for regulation given (information asymmetry, systemic externality, or deposit-insurance moral hazard) **AND** micro-prudential vs. macro-prudential correctly distinguished by both focus (individual vs. systemic) **AND** at least one example of a tool for each.
15–22	**15–22**	One clear rationale for regulation given, and the micro/macro distinction made correctly but without tool examples, or with the objective stated but not the focus.
8–14	**8–14**	Regulation rationale is generic ("banks need to be regulated to protect people") without a specific market-failure argument; micro/macro distinction attempted but muddled.
0–7	**0–7**	No credible rationale given, or micro-prudential and macro-prudential are used interchangeably with no distinction drawn.

The problem: moral hazard in an equity contract arises because the firm's managers (the **agent**) know more about their own effort and the firm's true prospects than outside shareholders (the **principal**) do, and managers bear only a fraction of the cost of poor decisions while shareholders bear the full downside — this is the **principal–agent problem**.

- **Monitoring (costly state verification):** shareholders, auditors, or delegated monitors such as venture capital firms incur the cost of verifying what management is actually doing, reducing the information gap that lets shirking or self-dealing go undetected.
- **Government regulation to increase information:** mandatory disclosure laws (audited financial statements, securities-law reporting requirements) force firms to release information they would not voluntarily disclose, lowering the cost of monitoring for every shareholder simultaneously.

- **Financial intermediaries acting as delegated monitors:** rather than every small shareholder duplicating the cost of monitoring, a venture capital firm or large institutional investor takes a concentrated equity stake (often with a board seat) and monitors on behalf of all investors, exploiting economies of scale in monitoring.
- **Using debt rather than equity where feasible:** a debt contract fixes the payment management owes regardless of how well the firm performs, which weakens management's incentive to overstate performance and reduces how closely investors need to monitor upside outcomes — this is part of why moral hazard is a specific argument for using debt finance over equity finance in some circumstances.

EXAM TECHNIQUE

Different textbooks and lecturers list this set slightly differently — some frame the fourth tool as "incentive/compensation contracts" (tying pay to firm performance) rather than the debt-vs-equity point above. Either is defensible; what earns marks is a correct *mechanism* for each tool, not the exact four-item list matching a specific slide verbatim.

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31–40	**31–40**	Four distinct tools given **AND** each explained with the correct mechanism by which it reduces moral hazard specifically (not just "it helps") **AND** the principal–agent framing is stated at the outset.
21–30	**21–30**	Four tools named with broadly correct mechanisms, but one or two explanations are thin or generic.
11–20	**11–20**	Only two or three tools given, or four tools listed with little to no explanation of mechanism.
0–10	**0–10**	Fewer than two credible tools given, or the answer confuses moral hazard with adverse selection throughout.

Question A2 Total: 100 Marks

Section A · Question 3

Bond pricing, time value of money, risk vs. uncertainty — 100 marks

A3(a) **[30 marks]**

A corporation has issued a bond with the following characteristics: Par value: $1,000. Time to maturity: 2 years. Coupon rate: 6 percent. The coupon payments are semi-annual. Calculate the price of this bond if the YTM is 5 percent.

A3(b) **[30 marks]**

Calculate the present value of an investment of €600 after 5 years at an interest rate of 5% per annum and at a compounding frequency of: Annual · Semi-annual · Quarterly.

A3(c) **[40 marks]**

What is the difference between Risk and Uncertainty? Use examples in your answer.

FULL MODEL ANSWER

Setting up the problem: with semi-annual coupons, a 2-year bond has **n = 4 periods**. The **semi-annual coupon** is the annual coupon rate applied to par and split in two: 6% × \$1,000 ÷ 2 = **\$30 per period**. The **semi-annual discount rate** is the YTM applied the same way: 5% ÷ 2 = **2.5% per period**. Using the bond-pricing formula from the exam's own formula sheet:

Formula: $P_B = \Sigma_{t=1}^{T} C_t / (1+r)^t + \text{ParValue} / (1+r)^T$, with C = \$30, r = 2.5%, T = 4 periods.

Period (t)	Cash flow (\$)	Discount factor $(1.025)^{-t}$	PV of cash flow (\$)
1	30.00	0.9756	29.27
2	30.00	0.9518	28.55
3	30.00	0.9286	27.86
4	1,030.00 (coupon + par)	0.9060	933.13
Bond price			**\$1,018.81**

Bond Price–Yield Relationship (2-Year, 6% Semi-Annual Coupon)

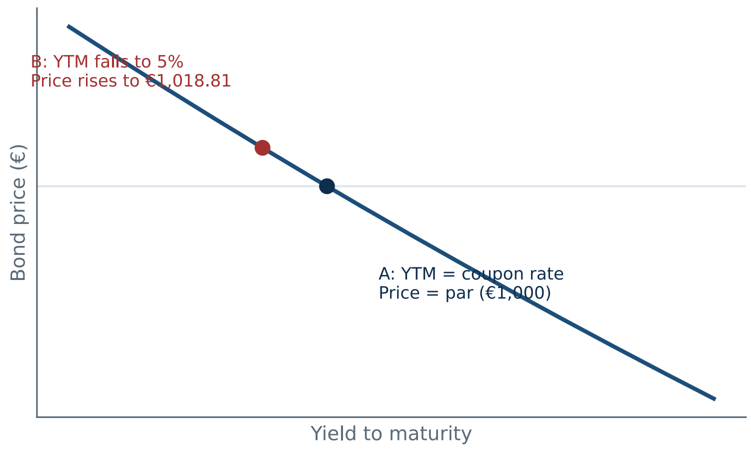

Fig. 3 — Bond price–yield relationship: at YTM = 6% (equal to the coupon rate) the bond prices exactly at par; as YTM falls to 5% the price rises above par to \$1,018.81, illustrating the inverse relationship between yield and price.

Result: the bond prices **above par** (\$1,018.81 > \$1,000) because its 6% coupon rate exceeds the 5% yield the market currently demands — investors are willing to pay a premium to lock in a coupon higher than the going rate.

EXAM TECHNIQUE

A very common slip is discounting at the **annual** rates (6% coupon, 5% YTM) over 2 periods instead of converting both to semi-annual and using 4 periods — since the question states coupons are paid semi-annually, both the coupon and the discount rate must be halved and the period count doubled. Under the own-figure rule, a fully consistent semi-annual calculation with one carried-through arithmetic slip still earns close to full marks; a calculation left on an annual basis throughout does not, since it uses the wrong method for the question asked.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	Correctly converts to 4 semi-annual periods, $30 coupon, and 2.5% discount rate **AND** correctly discounts all four cash flows **AND** arrives at $1,018.81 (or a consistent own-figure result from a single carried-through slip).
15–22	**15–22**	Correct semi-annual setup (periods, coupon, rate) but an arithmetic error in the discounting that is not carried through consistently, or one cash flow (commonly the final period's par value) omitted.
8–14	**8–14**	Bond-pricing formula applied on an annual basis (2 periods, $60 coupon, 5% rate) — wrong method for a semi-annual bond, but formula mechanically applied correctly.
0–7	**0–7**	No coherent application of the bond-pricing formula, or par value omitted from the final period entirely.

Setting up the problem: present value discounts a future sum back to today; the compounding frequency changes how many periods the €600 is discounted over and what rate applies each period, using PV = $C_t / (1 + r/m)^{mt}$, where m is the number of compounding periods per year.

Compounding	Periods (m×t)	Rate per period	PV of €600 (5 yrs, 5% p.a.)
Annual	5	5.000%	€470.12
Semi-annual	10	2.500%	€468.72
Quarterly	20	1.250%	€468.01

Result and why it moves this way: the present value **falls** as compounding gets more frequent (€470.12 → €468.72 → €468.01), because more frequent compounding raises the **effective** annual rate above the quoted 5% nominal rate, and a higher effective discount rate always reduces the present value of a fixed future sum. This is the mirror image of the future-value case: more frequent compounding grows a sum *faster* going forward, so it must discount a fixed future sum *more* going backward.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	All three PV figures correctly calculated with the correct rate/period adjustment for each compounding frequency **AND** the direction of the pattern (PV falling as compounding frequency rises) is correctly explained, not just observed.
15–22	**15–22**	At least two of the three PV figures correct with correct method shown; the pattern is stated but not explained, or one frequency's rate/period conversion is wrong.

8–14	**8–14**	Only the annual-compounding case correctly calculated; semi-annual and quarterly cases attempted with the annual rate/period incorrectly carried over.
0–7	**0–7**	PV formula not correctly applied for any of the three cases, or FV used where PV was asked for.

> **The distinction (Knight, 1921): risk** describes a situation where the possible outcomes are known and their **probabilities can be estimated** — from historical frequency, from theory, or from a well-understood random process. **Uncertainty** describes a situation where either the possible outcomes themselves, or the probabilities attached to them, **cannot be meaningfully estimated** at all.

- **Risk example:** the probability a diversified portfolio of investment-grade corporate bonds defaults in a given year can be estimated reasonably well from decades of historical default-rate data — an insurer or bank can price this and hold capital against it.
- **Uncertainty example:** the economic impact of a genuinely novel event — a first-of-its-kind pandemic, or an unprecedented geopolitical shock — has no comparable historical base rate to draw a probability distribution from; markets and models can only guess, not calculate.

> **How intermediaries respond to each:** financial intermediaries actively work to convert uncertainty into manageable risk — by pooling many independent, similar exposures (insurance-style diversification, which turns an individually uncertain outcome into a statistically predictable pool-level loss rate) and by building historical datasets and models that let a previously unmeasurable exposure be assigned a probability. This transformation is itself one of the core economic functions financial intermediaries perform (see Q. A1(c)) — it is precisely what makes a risk insurable or lendable against, where a raw uncertainty is not.

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31–40	**31–40**	Risk and uncertainty correctly distinguished on the basis of whether probabilities can be estimated **AND** a clear, well-chosen example given for each **AND** the answer explains how intermediaries convert uncertainty into risk (pooling/diversification, data/modelling).
21–30	**21–30**	Correct distinction with examples for both, but no discussion of how intermediaries manage the difference between the two.
11–20	**11–20**	Distinction attempted but imprecise (e.g. "uncertainty is just bigger risk") with at most one usable example.
0–10	**0–10**	Risk and uncertainty used interchangeably with no meaningful distinction drawn.

Question A3 Total: 100 Marks